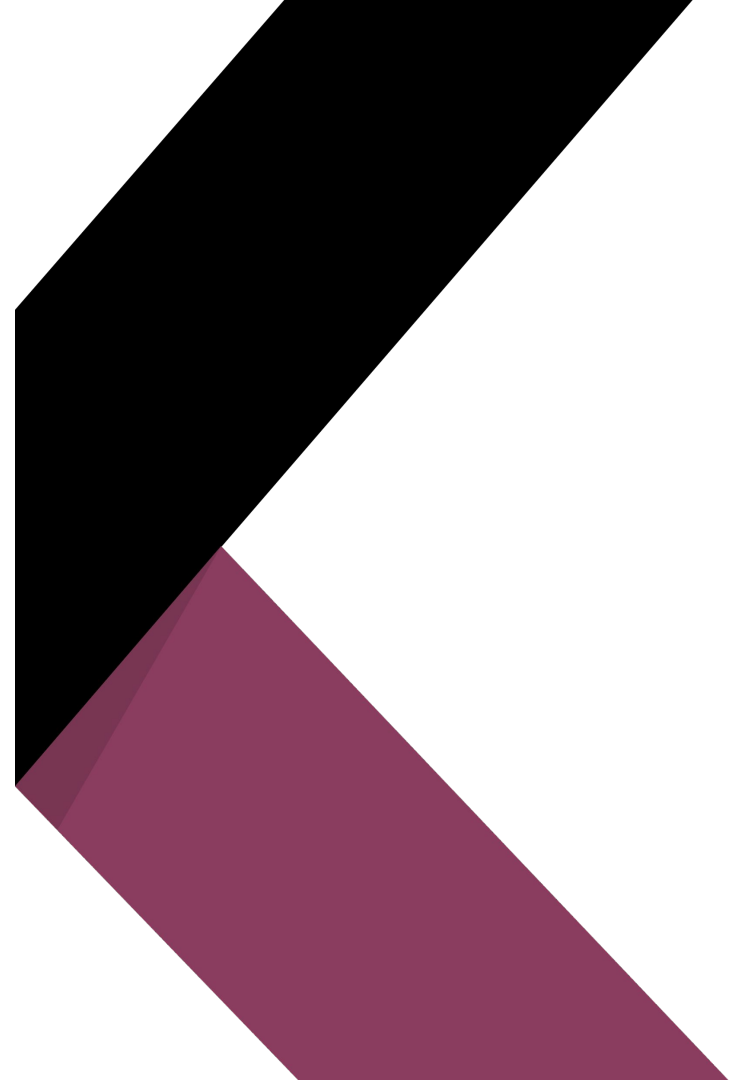


CANoe configuration

CANoe





CANoe



Config

CANoe

- CANoe differs from CANalyzer mainly in a use focused on databases
- Use Default to create a new configuration

The screenshot displays the CANoe Config interface. The top menu bar includes File, Home, Analysis, Simulation, Test, Diagnostics, Environment, Hardware, Tools, and Layout. The left sidebar contains a 'New' button (highlighted in red) and a list of options: Configuration Overview, Help, Sample Configurations, Options, Support, and Exit. The main area is titled 'Available Templates' and lists several categories of templates: Car2x (Car2x, Car2x Pilot PKI), Ethernet (ASR4_Ethernet, Ethernet, SOME-IP), FlexRay (ASR4_FlexRay, FlexRay, FlexRay Predefined CC Params), General (Default), and ISO11783 (ISO11783). The 'Default' template is selected. On the right, a panel shows configuration details for the 'Default' template, including File Name, Channels, Display Name, Category, and Description.

Available Templates

- Car2x
 - Car2x
 - Car2x Pilot PKI
- Ethernet
 - ASR4_Ethernet
 - Ethernet
 - SOME-IP
- FlexRay
 - ASR4_FlexRay
 - FlexRay
 - FlexRay Predefined CC Params
- General
 - Default
- ISO11783
 - ISO11783

Details of Template 'Default'

File Name:

Channels:

Display Name: Default

Category: General

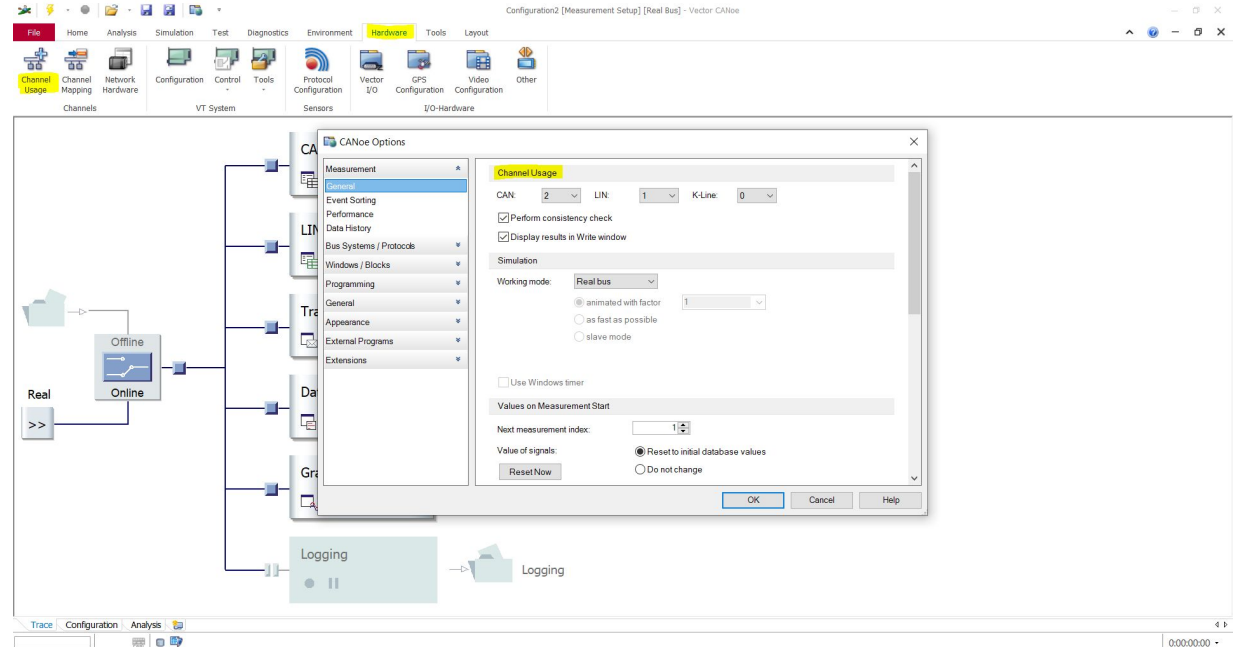
Description: Default configuration with channels configured for all activated bus systems



Config

CANoe

- The first step is the Hardware configuration, number of channels, baud rate and association of hardware channels with logical channels
- In Hardware->Channel Usage ->General We set the number of CAN and LIN channels

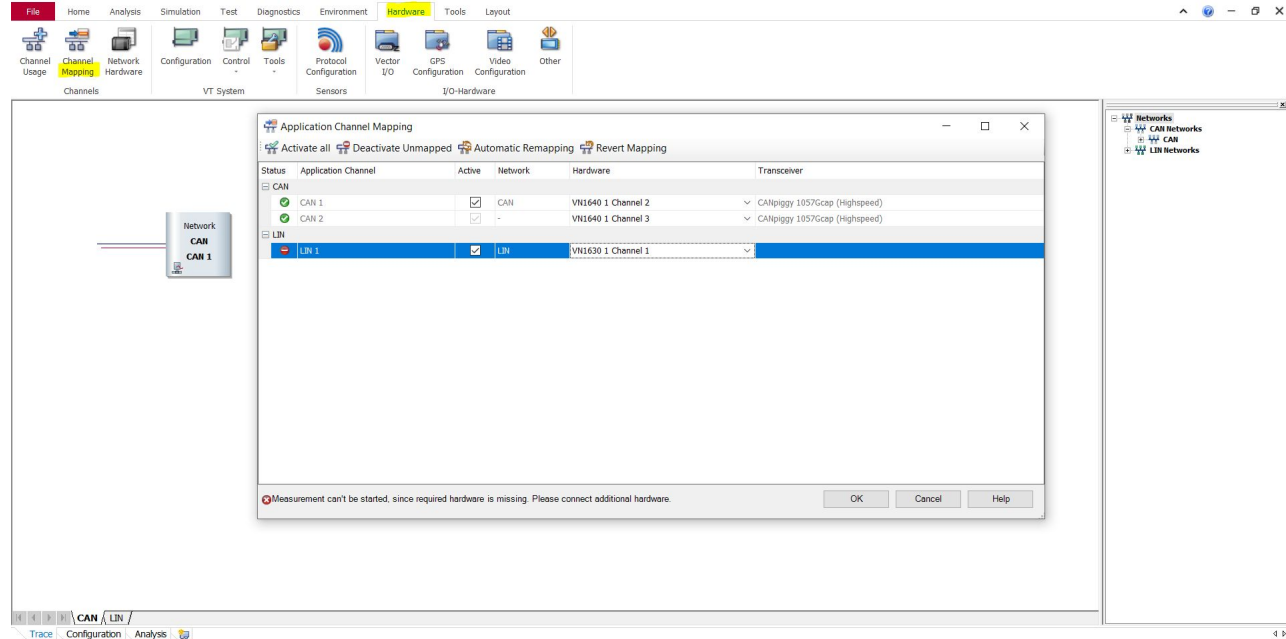




Config

CANoe

- In Hardware->Channel Mapping
- Association of Application Channel with the corresponding hw channel on vn

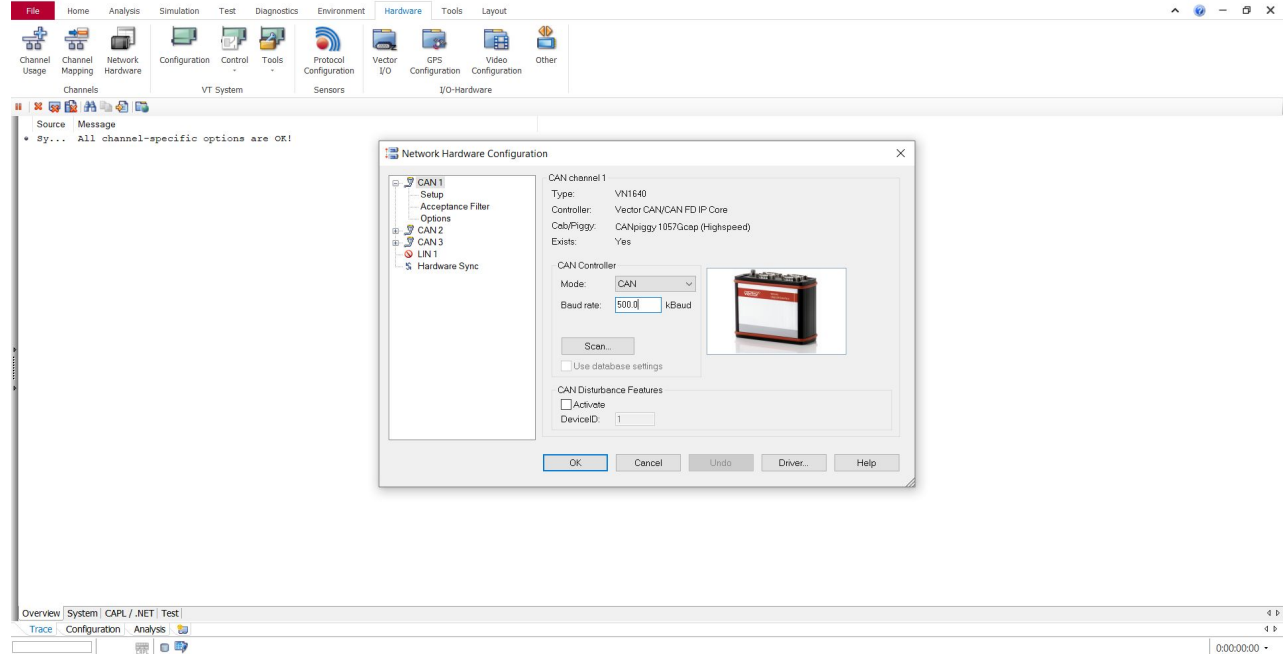




Config

CANoe

- In Hardware->Network Hardware
- we can change the baud rate for B=500 Kbaud BH=125 Ccan=500

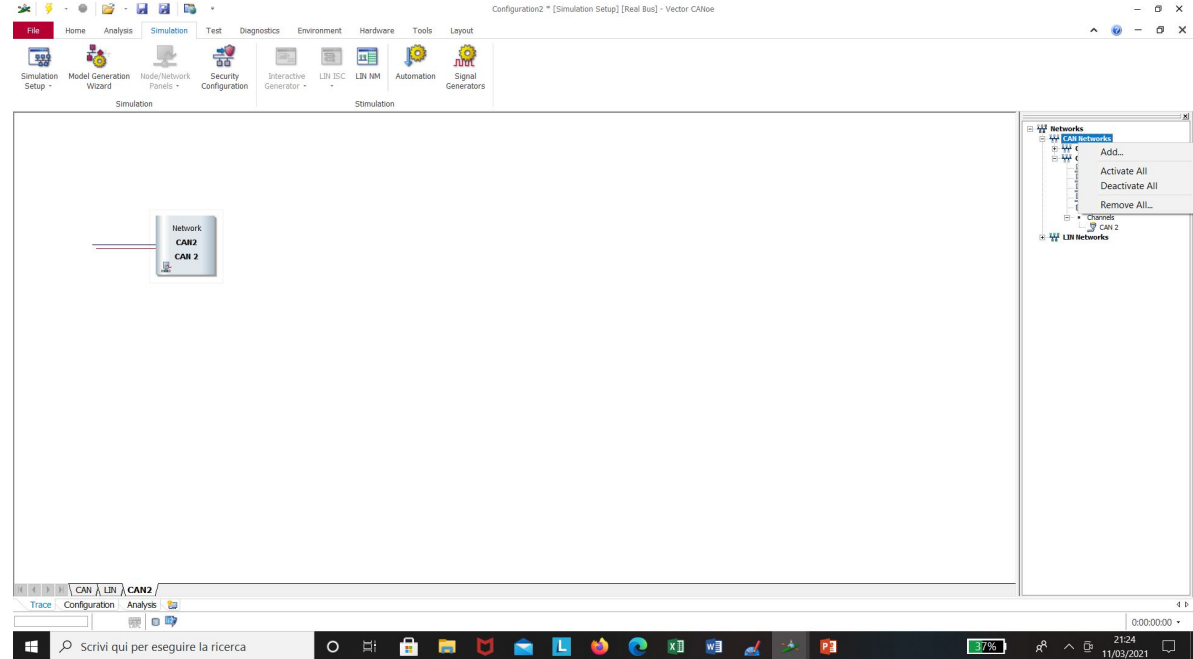




Config

CANoe

- In Simulation>Simulation Setup -> right click on CAN Network ->Add For new Network CAN
- right click on LIN Network->ADD For new Lin network
- We can create more CAN networks even if we don't have enough HW channels available
- The channels don't essential will have to be disabled in channel mapping

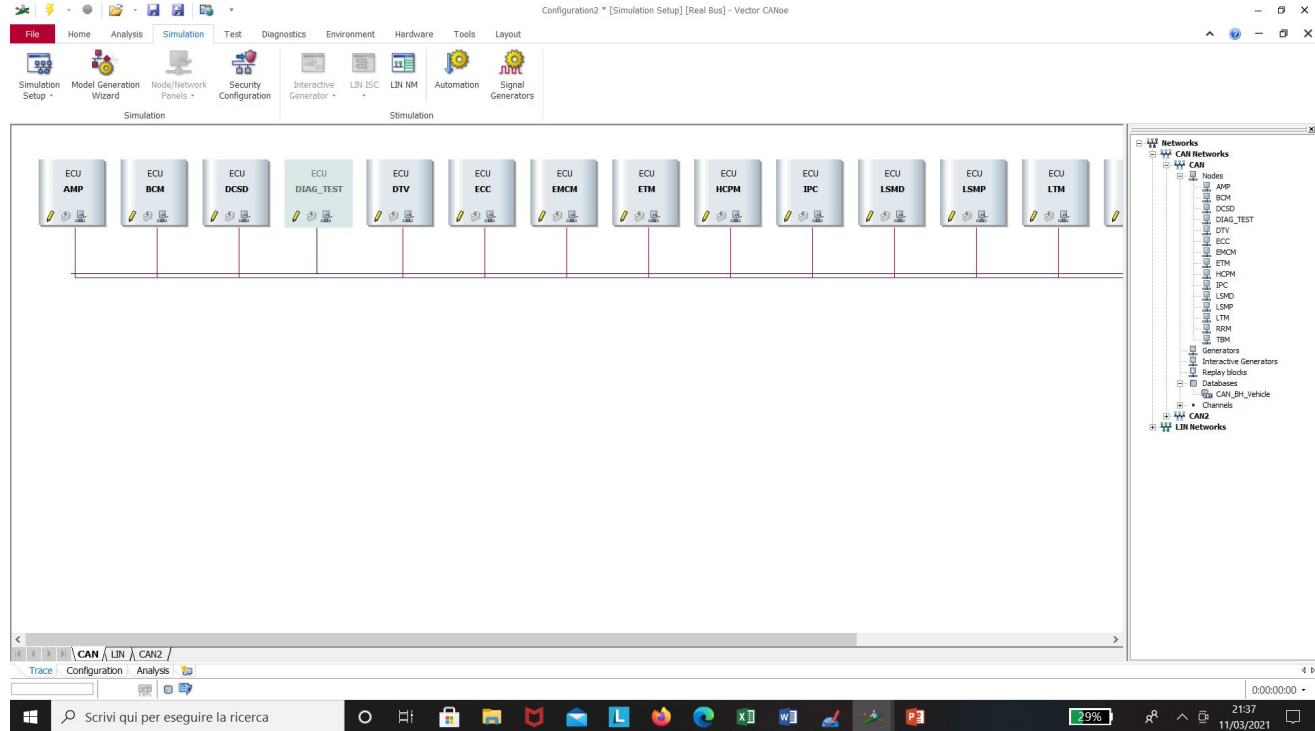




Config

CANoe

- After import we have all the simulated ECUs
- CANoe through the database simulates the CAN messaging of the ECU but not the logic
- select a node and press the sweeper bar to deactivate it

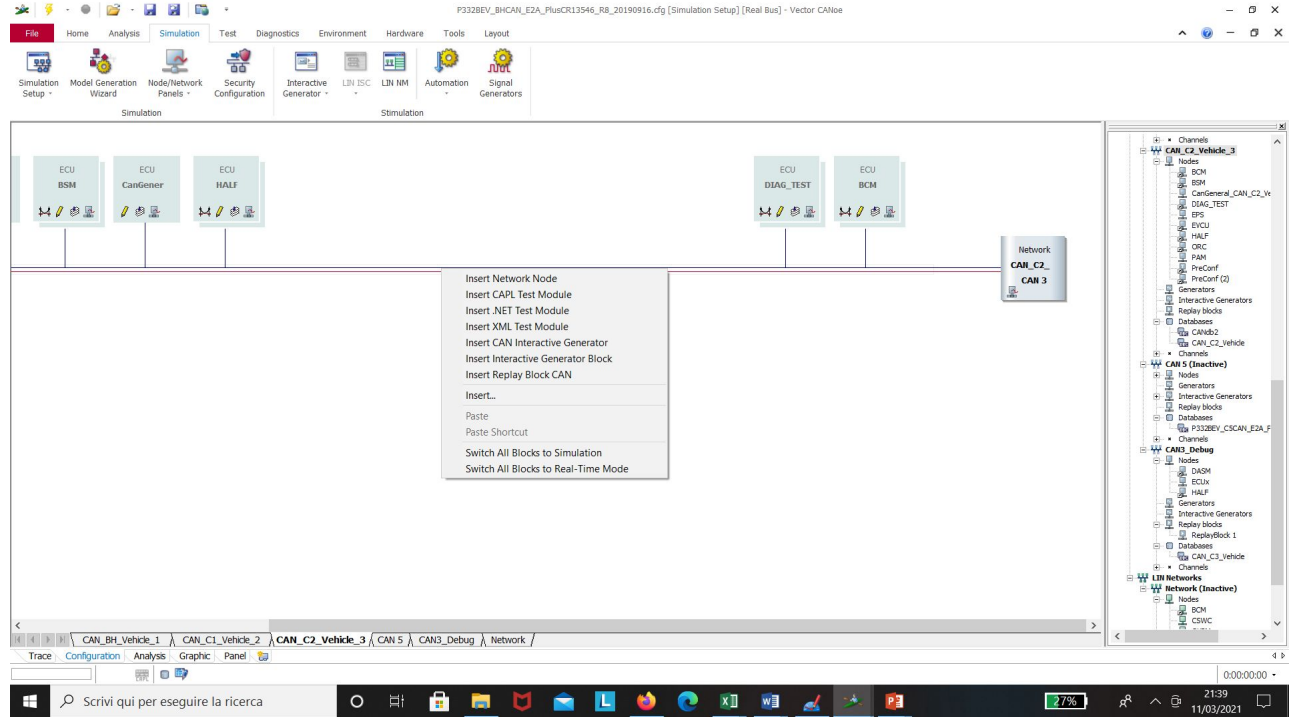




Config

CANoe

- to deactivate all nodes select Switch ALL blocks to real-time mode. Mainly when we are in the vehicle
- Right click on bus





Config

CANoe

- CANoe automatically send messages with default values, for change the value press the node panel and reload the message after change of the signal

The screenshot shows the Vector CANoe Configuration2 interface. At the top, there's a menu bar with options like File, Home, Analysis, Simulation, Test, Diagnostics, Environment, Hardware, Tools, and Layout. Below the menu is a toolbar with icons for Simulation Setup, Model Generation Wizard, Node/Network Panels, Security Configuration, Interactive Generator, LIN JSC, LIN NM, Automation, and Signal Generators. The main workspace displays a network diagram with several ECUs connected to a central bus. The ECUs include AMP, BCM, DCSD, DIAG_TEST, DTV, ECC, EMCM, ETM, HCPM, IPC, LSMD, LSMP, and LTM. The ECC node is highlighted with a red circle and labeled 'Node Panel'. On the right side, there's a 'Networks' panel showing a tree view of CAN networks and nodes. Below the network diagram, there's a 'Messages' panel showing a list of messages. The 'STATUS_CLIMATE' message is selected, and its details are displayed in a table. The table has columns for Signal, Physical, Raw Value, and Symbolic Value. The 'RHeatedWindowCmd' signal is highlighted in blue. On the far right, there's a 'Triggers' panel with an 'Update' button circled in red.

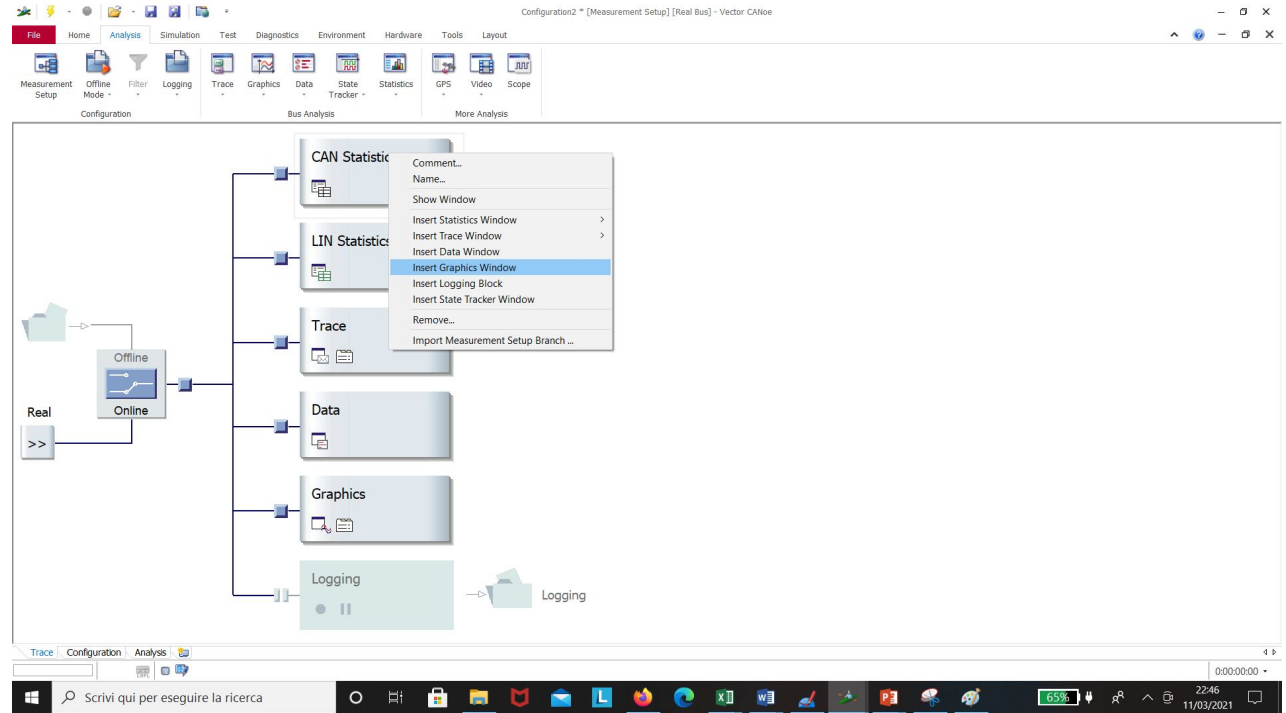
Signal	Physical	Raw Value	Symbolic Value
CompressorACReqSts	0	0x0	Not_Active
RHeatedWindowCmd	0	0x0	Not_Active
HVAC_Blower_Speed	0	0x0	-
MaxDefrostSts	0	0x0	Not_Active
PowerModeSts_Climate	0	0x0	Active
RHeatedWindowCmd	0	0x0	Not_Active
StopStartClimateInfo	0	0x0	SPST_ON
StopStartEnable	0	0x0	False



Config

CANoe

- The tab Measurement Setup is very similar to that of CANalyzer the difference is that program node and IG block are added in Simulation Setup





Config

CANoe

- In Graphic let's add the signal to have a graphic display

The screenshot displays the CANoe software interface with the 'Configuration2' project open. The 'Graphics' tab is active, showing a signal trace. A context menu is open over the trace, with options: 'Add Signals...' (Ctrl+I), 'Add Variables...' (Ctrl+E), 'User Defined Signals' (indicated by a right arrow), 'Configuration...' (Ctrl+T), 'Paste' (Ctrl+V), and 'Y-Axis Left/Right'. A 'Symbol Selection' dialog box is also open, showing a tree view of the project structure. The 'Tx Frames' folder is expanded, and the 'STATUS_CLIMATE' frame is selected. The 'Channel Mask' column shows the bit positions for the signals. The 'Comment' column provides additional information about the signals.

Name	Channel Mask	Start Bit	Comment
COMPRESSOR_ACTIVATION	ECC	54	Used to indicate the...
WINDSHIELD_WASHER	ECC	62	Used to indicate the...
WINDSHIELD_WASHER_SPEED	ECC	54	Used to indicate the...
WINDSHIELD_WASHER_STATUS	ECC	54	Used to indicate the...
WINDSHIELD_WASHER_TEMP	ECC	62	Used to indicate the...

Frame name: STATUS_CLIMATE
Start bit: 54
Comment: Used to indicate the user request of compressor activation

Buttons: OK, Cancel, Apply, Help



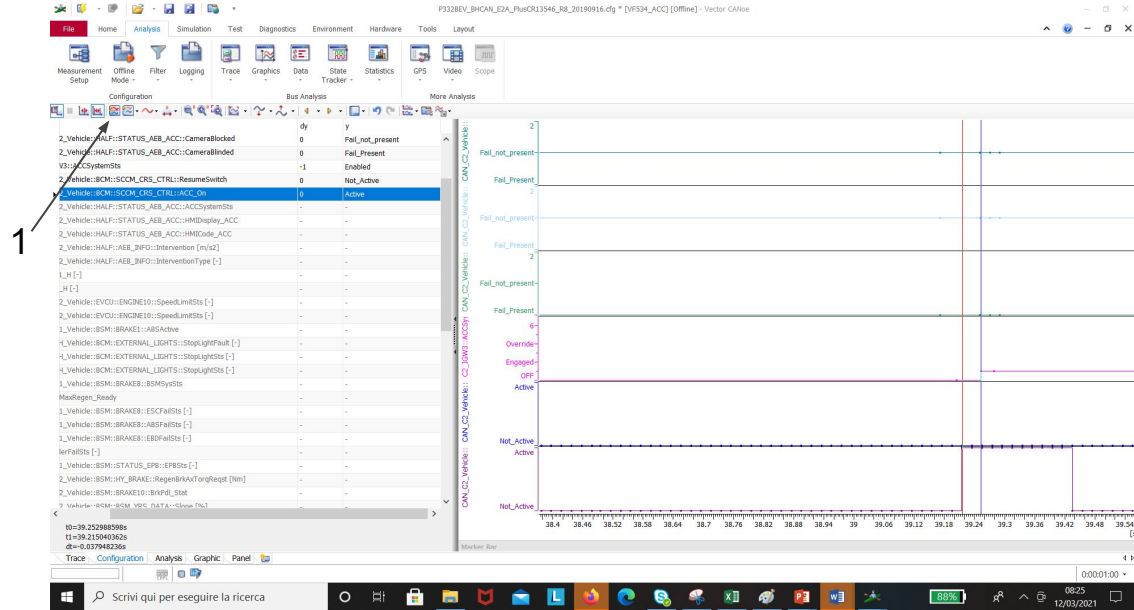
-



Config

CANoe

- Using the button indicated with 1, it is possible to measure the time between two samples. Example of ACC activation key and HALF response

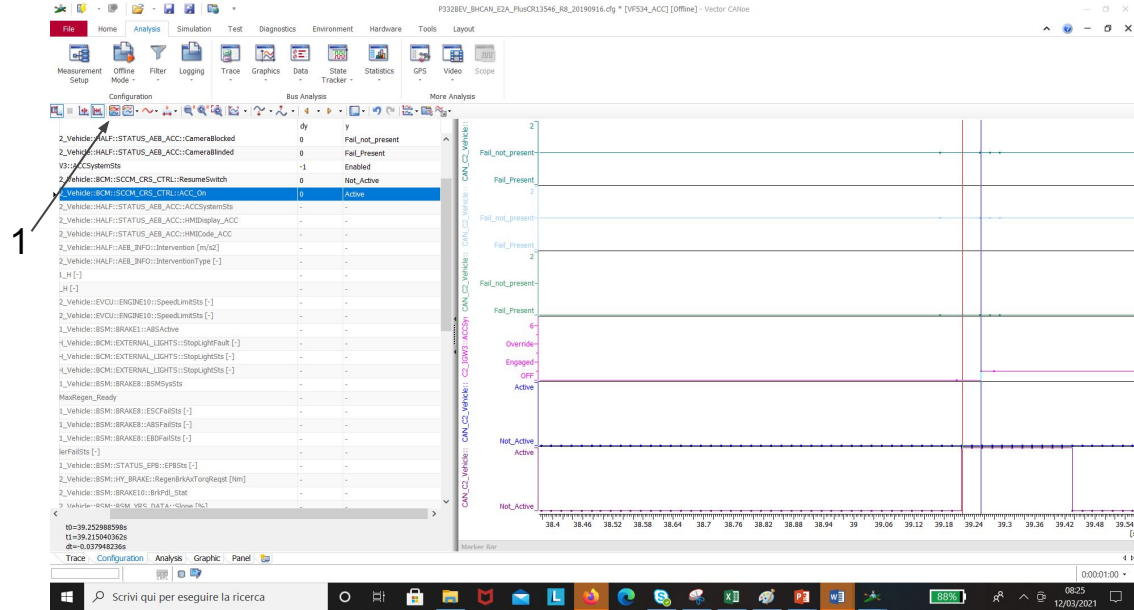




Config

CANoe

- Using the button indicated with 1, it is possible to measure the time between two samples. Example of ACC activation key and HALF response

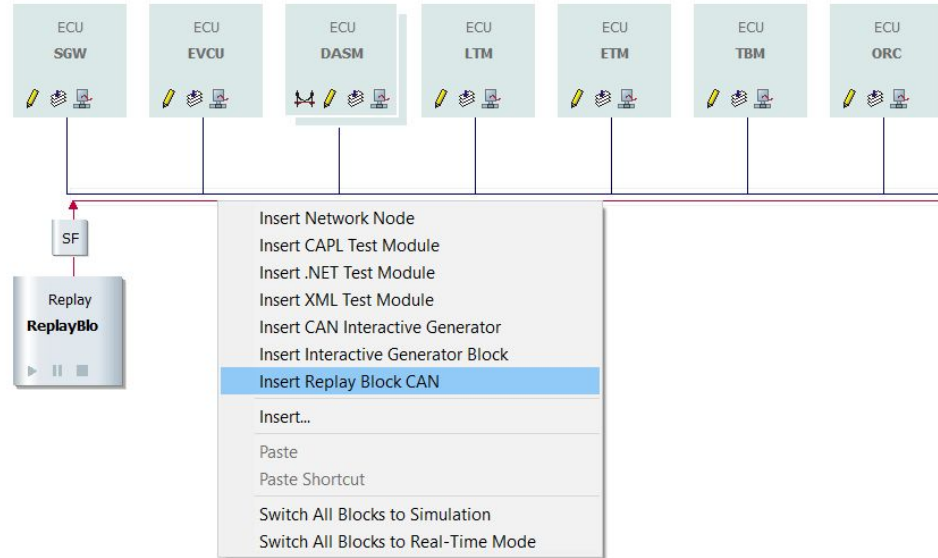




Config

CANoe

- During testing of individual components it is useful to replicate the log capture in the car, to control behavior stand alone.
- Insert replay block CAN and insert the filter to block the messages in the log of the ECU we are testing



CONTACTS

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